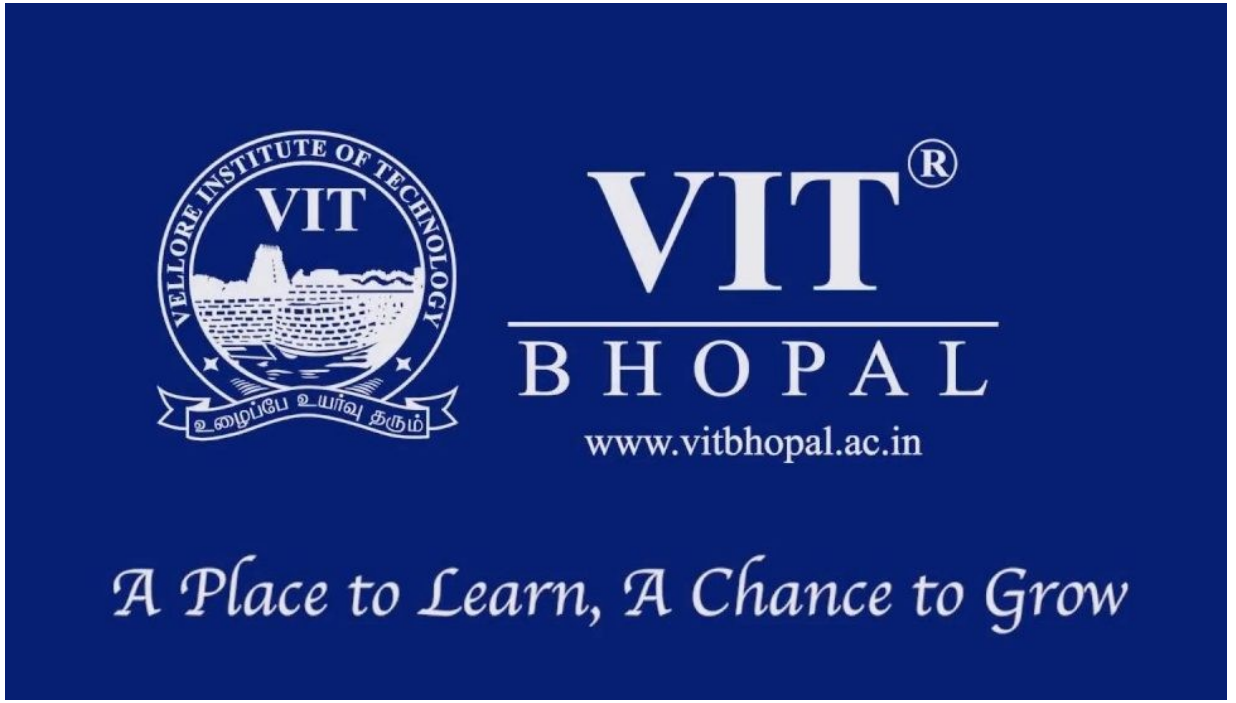


Introduction

Dr. Saurav Prasad

Overview of the course

- Course Name: Introduction to Computational Chemistry
- Course Code: CHY1005
- LTP Course: Lecture, Tutorial, and Practical
- 4 Credits; 3 Sessions per week

Course structure

- **Module 1: Introduction:** Overview of the course, history and promises of computational chemistry, tools for computational chemistry, units, errors in computed quantities.
- **Module 2: Quantum Chemistry:** Historical development, Bohr's atomic model, de Broglie wavelength, Heisenberg uncertainty principle, Schrodinger equation, Wave function, particle in a box, Hydrogen atom, radial and angular solution to hydrogen atom, applications and limitations.
- **Module 3: Thermodynamics:** Intensive and extensive variables, state and path functions, Laws of Thermodynamics (First law and enthalpy; second law and entropy, spontaneity, and equilibrium; third law and absolute entropy) free energy, Gibbs and Maxwell's relations, Ideal and real gases.

Course structure

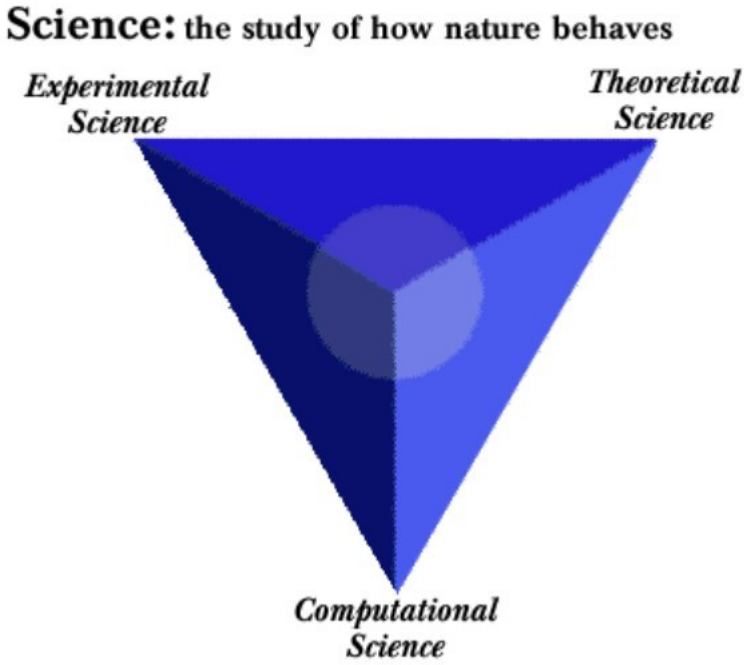
- **Module 4: Potential Energy Surfaces:** Chemical bonds and intermolecular interactions, Types of intermolecular interactions (charge distribution of isolated molecules, electrostatic interaction, induction interaction, London or dispersion forces, hydrogen bonding, repulsive interaction, relative contribution of different terms), representing the potential energy surfaces (pair additivity, rare gas), intramolecular interactions (bond stretching, angle bending, torsional and improper terms).

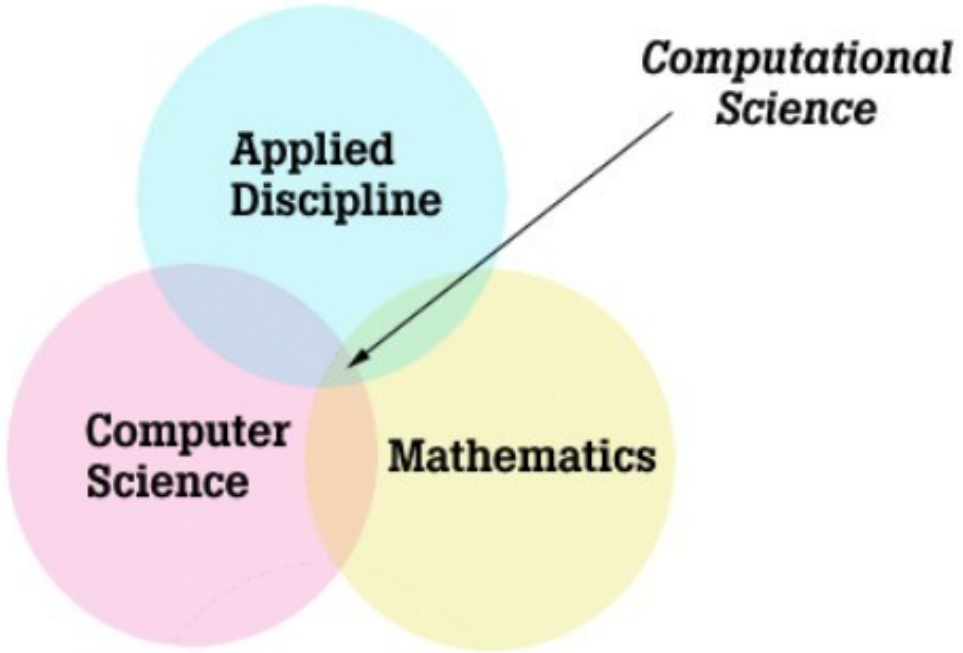
- **Module 5: Molecular Dynamics:** Introduction to ensembles, force fields, integration of Newton's laws of motion, force calculation, energy minimization, periodic boundary conditions, choice of input configuration, velocities, and time-step, applications, and calculation of simple thermodynamic variables.

- **Guest Lectures**

Overview of Computational Science

- **Computational science** is the application of computational and numerical techniques to solve large and complex problems.



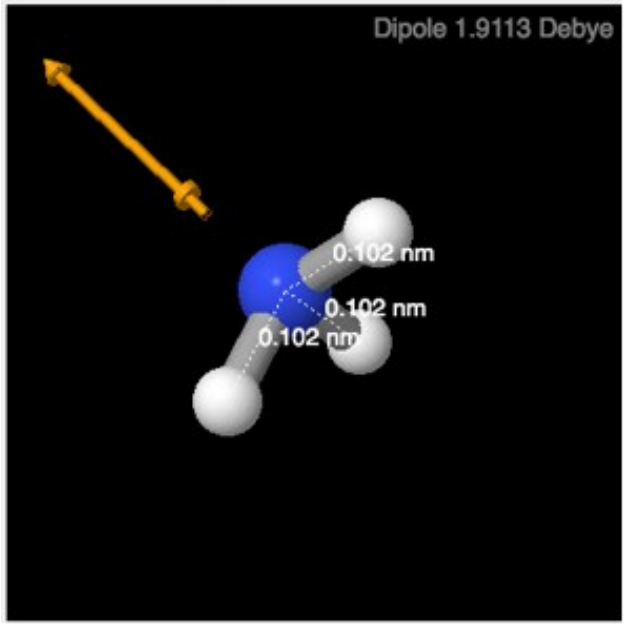


Overview of Computational Chemistry

- **Computational chemistry** is simply the application of chemical, mathematical and computing skills to the solution of interesting chemical problems.
- Useful way to investigate materials that are too difficult to find or too expensive to purchase.
- Make predictions before running the actual experiments.
- Schrödinger equation is the basis.

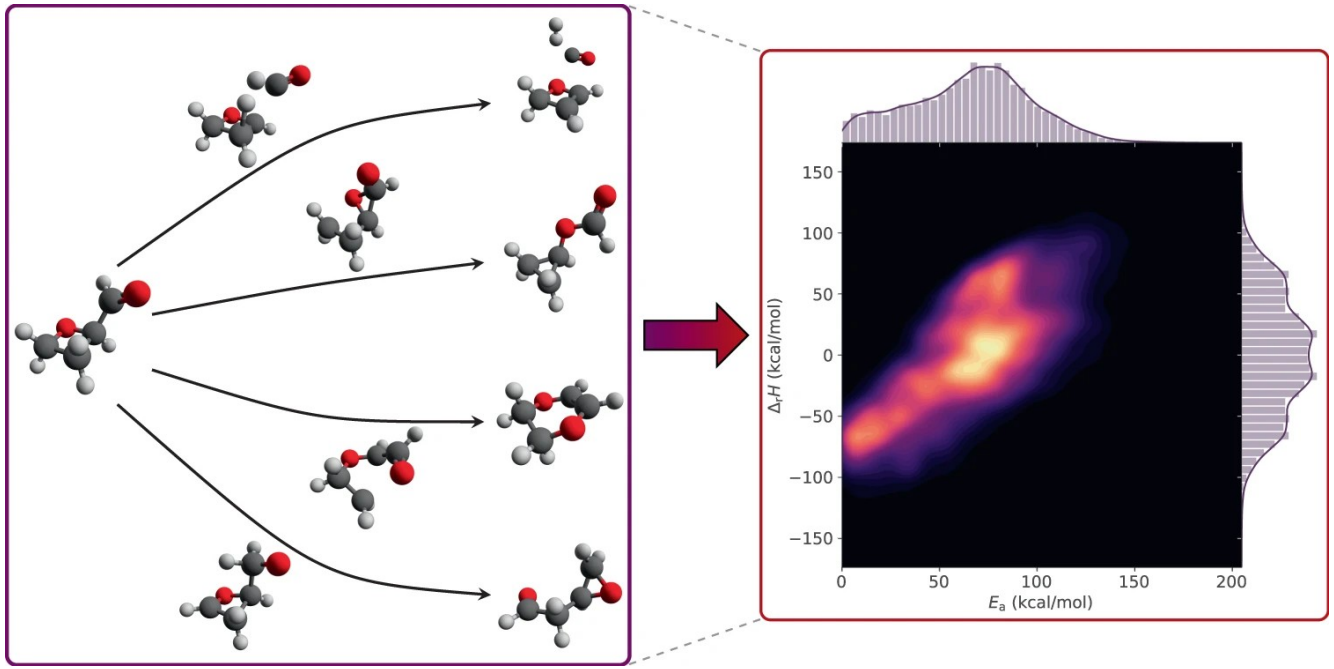
Questions commonly investigated computationally are

- **Molecular geometry:** the shapes of molecules – bond lengths, angles and dihedrals.



- **Energies of molecules and transition states:** this tells us which isomer is favored at equilibrium, and (from transition state and reactant energies) how fast a reaction should go.

Questions commonly investigated computationally are

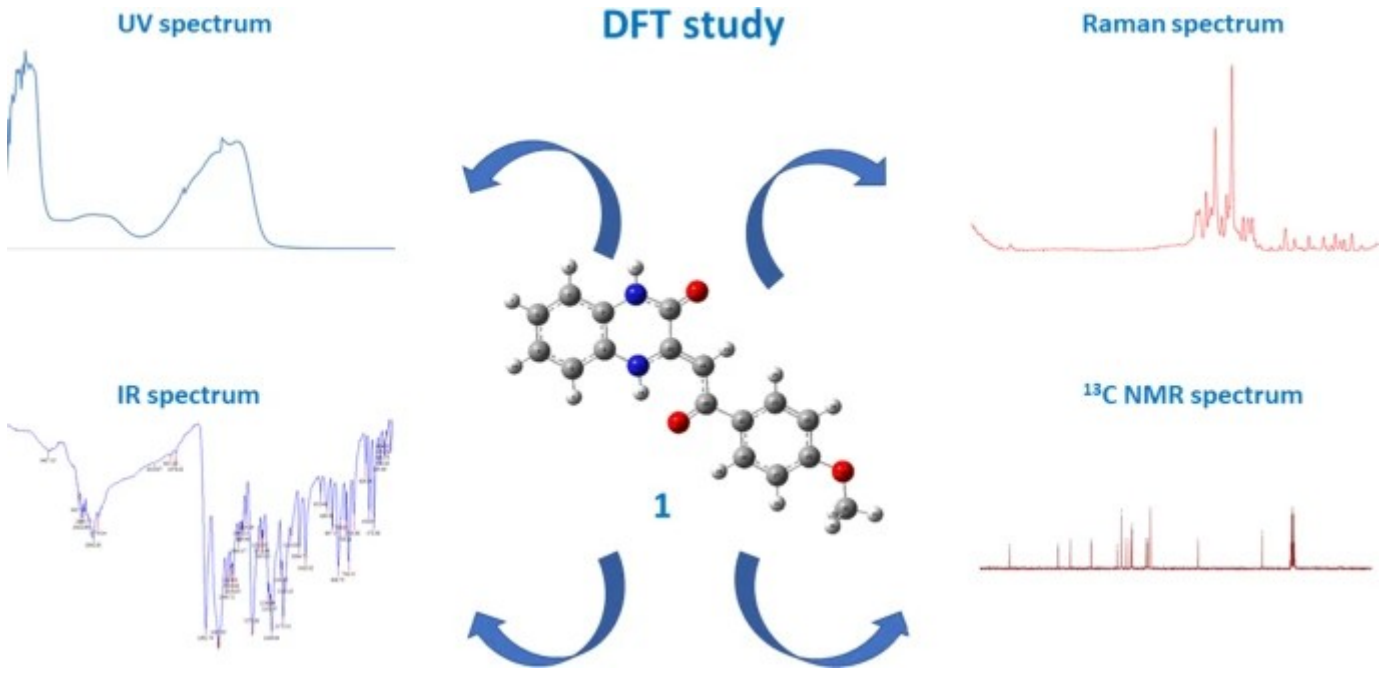


Grambow, C.A., Pattanaik, L. & Green, W.H. Reactants, products, and transition states of elementary chemical reactions based on quantum chemistry. *Sci Data* **7,** 137 (2020).

- **Chemical reactivity:** for example, knowing where the electrons are concentrated (nucleophilic sites) and where they want to go (electrophilic sites) enables us to predict where various kinds of reagents will attack a molecule.

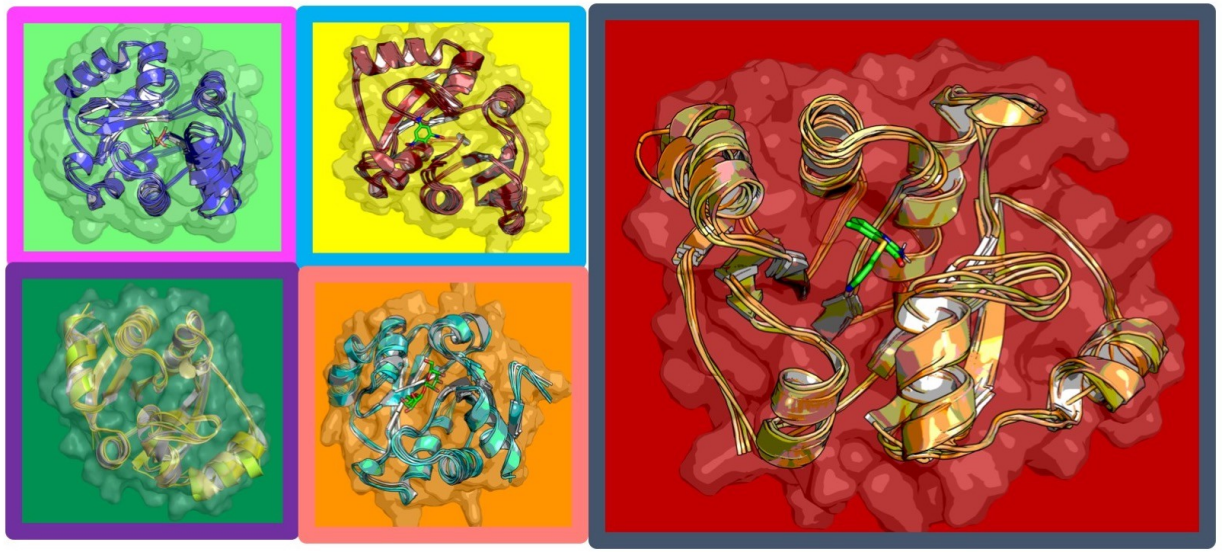
Questions commonly investigated computationally are

- **IR, UV and NMR spectra:** these can be calculated, and if the molecule is unknown, someone trying to make it knows what to look for.



Questions commonly investigated computationally are

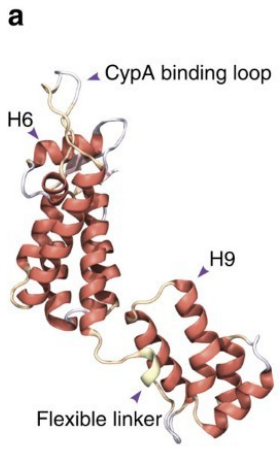
- **The interaction of a substrate with an enzyme:** seeing how a molecule fits into the active site of an enzyme is one approach to designing better drugs.

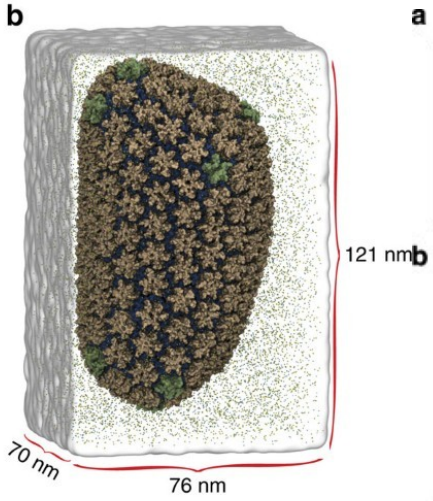


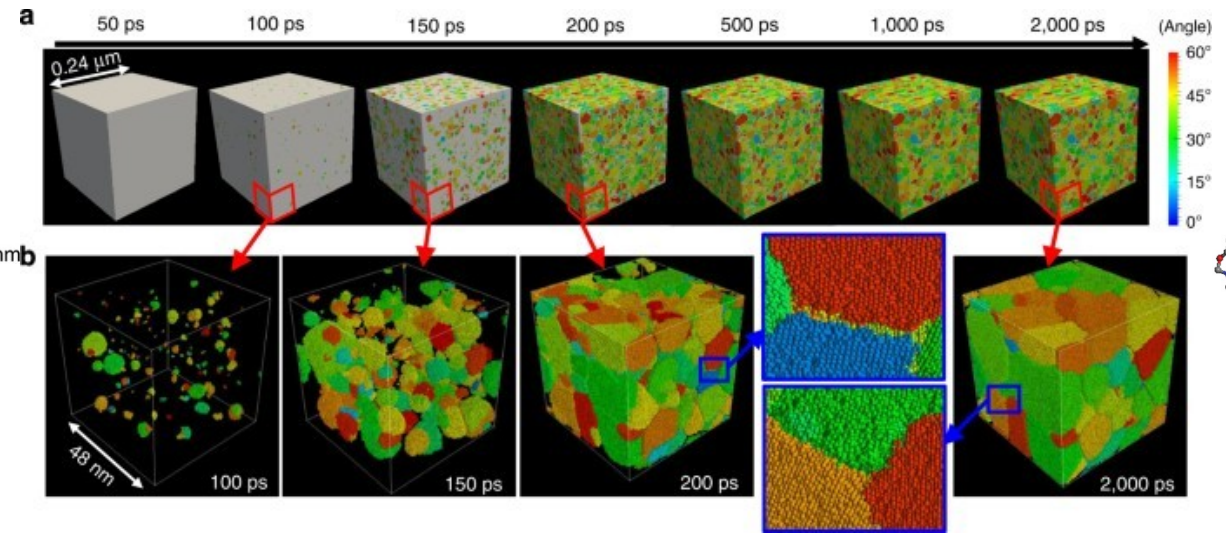
Drug discovery to fight COVID-19: Deep learning paired with drug docking and molecular dynamics simulations.

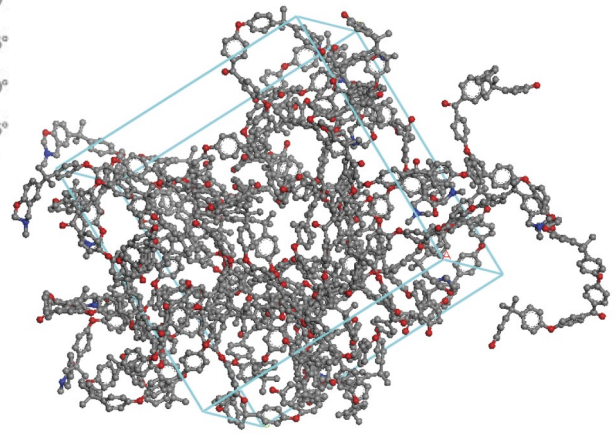
Questions commonly investigated computationally are

- **The physical properties of substances:** these depend on the properties of individual molecules and on how the molecules interact in the bulk material. For example, the strength and melting point of a polymer (e.g. a plastic) depend on how well the molecules fit together and on how strong the forces between them are.









Tools of Computational Chemistry

- Two approaches (general):

(a) **Quantum Mechanical Methods:** numerical computation of molecular electronic structures by density functional methods, *ab initio* and semi-empirical techniques.

(b) **Classical Computational Methods:** formulation of analytical expressions for the properties of molecules and their reactions. Molecular Mechanics/Molecular Dynamics/Monte – Carlo Methods.

Tools of Computational Chemistry

Method Type		Advantages	Disadvantages	Best for
Molecular Mechanics	• uses classical physics • relies on force-field with embedded empirical parameters	• Computationally least intensive - fast and useful with limited computer resources • can be used for molecules as large as enzymes	• particular force field applicable only for a limited class of molecules • does not calculate electronic properties • requires experimental data (or data from *ab initio*) for parameters	• large systems (thousands of atoms) • systems or processes with no breaking or forming of bonds
Semi-Empirical	• uses quantum physics • uses experimentally derived empirical parameters • uses approximation extensively	• less demanding computationally than *ab initio* methods • capable of calculating transition states and excited states	• requires experimental data (or data from *ab initio*) for parameters • less rigorous than *ab initio*) methods	• medium-sized systems (hundreds of atoms) • systems involving electronic transitions
Ab Initio	• uses quantum physics • mathematically rigorous, no empirical parameters • uses approximation extensively	• useful for a broad range of systems • does not depend on experimental data • capable of calculating transition states and excited states	• computationally expensive	• small systems (tens of atoms) • systems involving electronic transitions • molecules or systems without available experimental data ("new" chemistry) • systems requiring rigorous accuracy

Hybrid Methods: QM/MM

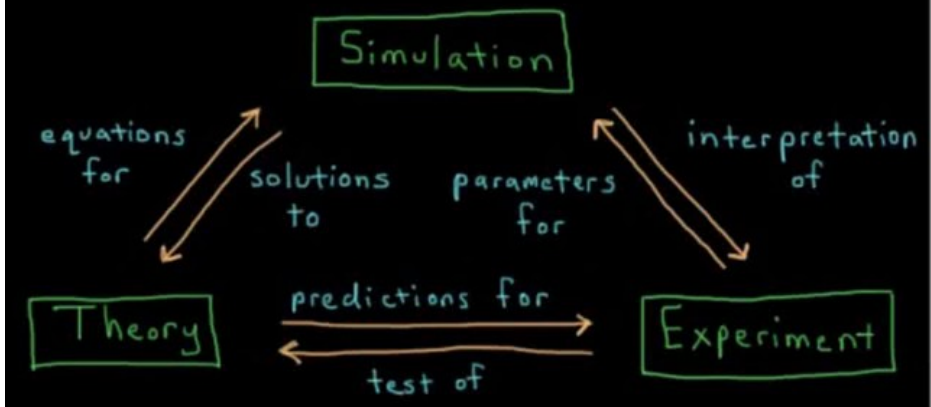
Tools of Computational Chemistry

- ***Ab initio* methods** – based on approximate solutions of the Schrodinger equation without appeal to fitting to experiment.
- **Semiempirical methods** – based on approximate solutions of the Schrodinger equation with appeal to fitting to experiment (i.e. using parameterization).
- **Density functional theory (DFT) methods** – based on approximate solutions of the Schrodinger equation, bypassing the wavefunction that is a central feature of ab initio and semiempirical methods.
- **Molecular mechanics** – based on a ball-and-springs model of molecules.
- **Molecular dynamics methods** study molecules in motion.

Revision: Introduction to Computational Chemistry

Introduction to Computational Chemistry

The use of computers to solve the equations of a theory or model for the properties of chemical systems



Model: simplified representation of a physical system

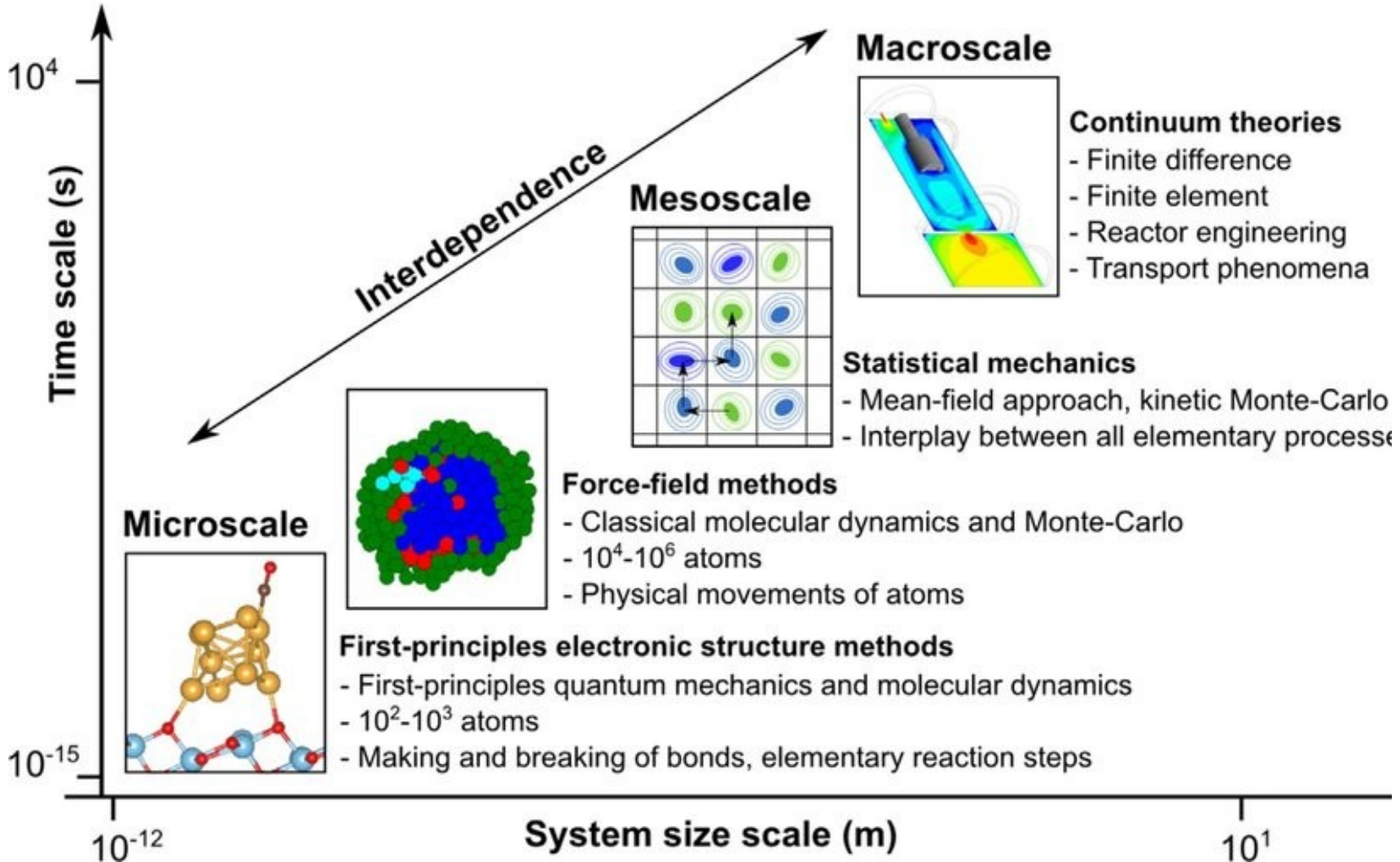
Computing much faster now

- Moore's Law
- parallel computing
- cheaper hardware
- better software

Properties include:

- energies
- structures
- spectra

Length and time scales



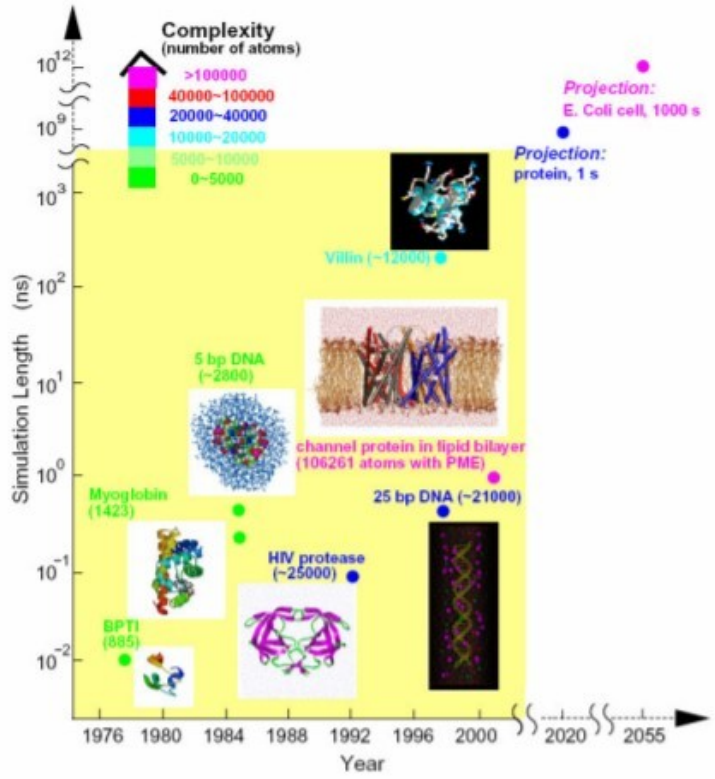


Figure 1.2. The evolution of molecular dynamics simulations with respect to simulation lengths (see also Table 1.2 and Figure 1.1). The data points for 2020 and 2055 represent extrapolations from the 1977 BPTI [136] and 1998 villin [57, 55] simulations, assuming a computational power increase by a factor of 10 every 3–4 years, as reported in [56].

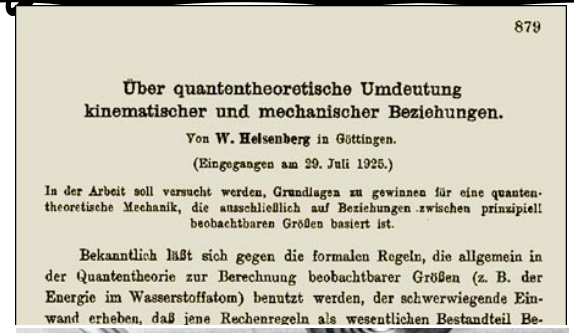
Conducting a computational project

- These questions should be answered
  - What do you want to know?
  - How accurate does the prediction need to be?
  - How much time can be devoted to the problem?
  - What approximations are being made?

- The answers to these questions will determine the type of calculation, model to be used.

History and evolution of Computational Chemistry

- **1925:** Heisenberg publishes his first paper on quantum mechanics (Z. Phys., 1925, 33, 879).
- **1926:** Schrödinger publishes his first paper on the theory of quantum mechanics (Ann. Phys., 1926, 79, 361).
- **1931:** Pi electron theory postulated by Hückel (Z. Phys., 1931, 70, 204).
- **1943:** The first computer, the ENIAC (Electronic Numerical Integrator and Computer) is built for the US Army Ordnance Department.
- **1951:** The first UNIVAC (Universal Automatic Computer) is delivered to the Census Bureau.
- **1953:** Metropolis and co-workers describe the application of the Monte Carlo method of simulation to physical chemistry problems (J. Chem. Phys., 1953, 1087, 21).



879

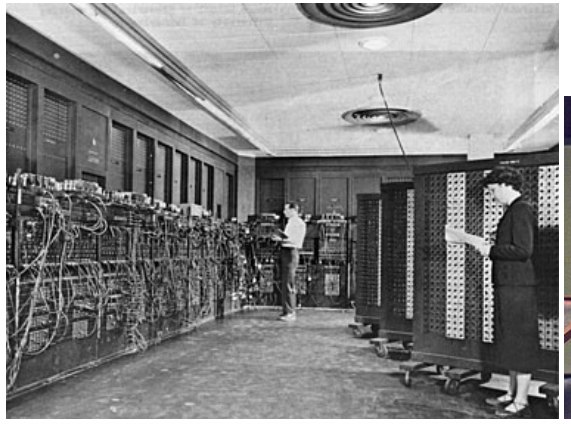
Über quantentheoretische Umdeutung kinematischer und mechanischer Beziehungen.

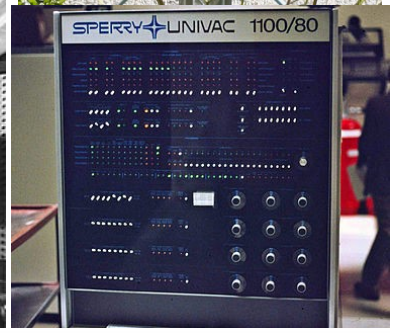
Von **W. Heisenberg** in Göttingen.

(Eingegangen am 29. Juli 1925.)

In der Arbeit soll versucht werden, Grundlagen zu gewinnen für eine quantentheoretische Mechanik, die ausschließlich auf Beziehungen zwischen prinzipiell beobachtbaren Größen basiert ist.

Bekanntlich läßt sich gegen die formalen Regeln, die allgemein in der Quantentheorie zur Berechnung beobachtbarer Größen (z. B. der Energie im Wasserstoffatom) benutzt werden, der schwerwiegende Einwand erheben, daß jene Rechenregeln als wesentlichen Bestandteil Be-





THE JOURNAL OF CHEMICAL PHYSICS VOLUME 21, NUMBER 6 JUNE, 1953

Equation of State Calculations by Fast Computing Machines

NICHOLAS METROPOLIS, ARIANNA W. ROSENBLUTH, MARSHALL N. ROSENBLUTH, AND AUGUSTA H. TELLER,
Los Alamos Scientific Laboratory, Los Alamos, New Mexico

AND

EDWARD TELLER,* *Department of Physics, University of Chicago, Chicago, Illinois*

(Received March 6, 1953)

A general method, suitable for fast computing machines, for investigating such properties as equations of state for substances consisting of interacting individual molecules is described. The method consists of a modified Monte Carlo integration over configuration space. Results for the two-dimensional rigid-sphere system have been obtained on the Los Alamos MANIAC and are presented here. These results are compared to the free volume equation of state and to a four-term virial coefficient expansion.

History and evolution of Computational Chemistry

- **1955:** Scherr reports the first ab initio calculation for a large system, N_2.
- **1956:** Alder and Wainwright –molecular dynamics (MD) simulation of hard spheres.
- **1957:** Pople publishes details on the application of self-consistent molecular orbital methods to pi electrons (J. Phys. Chem., 1957, 61, 6).
- **1958:** The first integrated circuit board is constructed by Jack Kilby at Texas Instruments.
- **1961:** Hendrickson publishes the results of calculations of relative conformational stabilities of cyclohexane (J. Amer. Chem. Soc., 1961, 83, 5537).
- **1964:** Hansch and Fujita describe a new approach to analyzing drug actions: QSAR, a quantitative structure activity relationship (Hansch, C and Fujita, T., J. Amer. Chem. Soc., 1964, 86, 1616).
- **1964:** Rahman–MD simulation of liquid Ar

THE JOURNAL OF CHEMICAL PHYSICS VOLUME 31, NUMBER 2 AUGUST, 1959

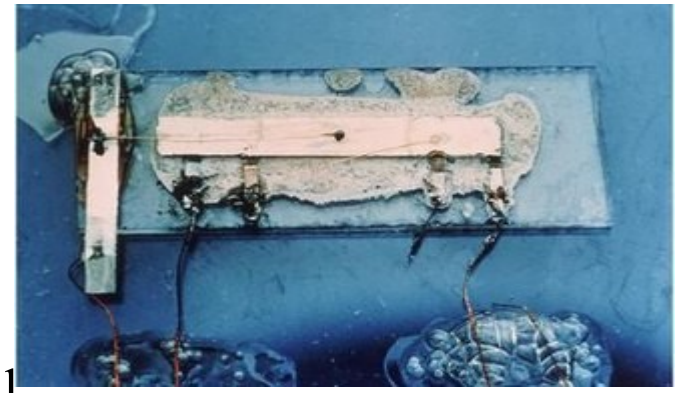
Studies in Molecular Dynamics. I. General Method*

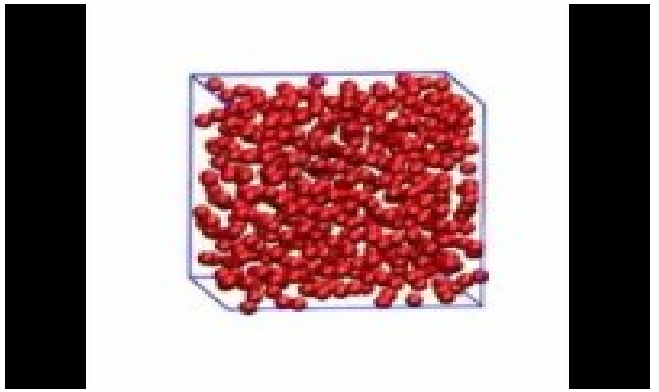
B. J. Alder and T. E. Wainwright

Lawrence Radiation Laboratory, University of California, Livermore, California

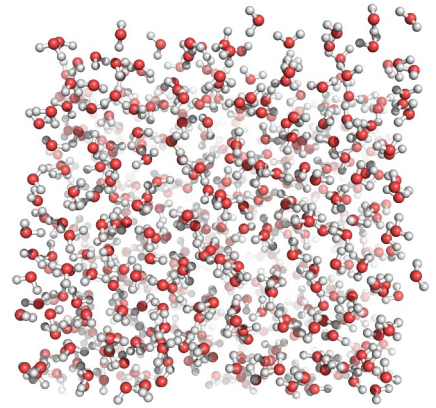
(Received February 19, 1959)

A method is outlined by which it is possible to calculate exactly the behavior of several hundred interacting classical particles. The study of this many-body problem is carried out by an electronic computer which solves numerically the simultaneous equations of motion. The limitations of this numerical scheme are enumerated and the important steps in making the program efficient on the computers are indicated. The applicability of this method to the solution of many problems in both equilibrium and nonequilibrium statistical mechanics is discussed.





History and evolution of Computational Chemistry

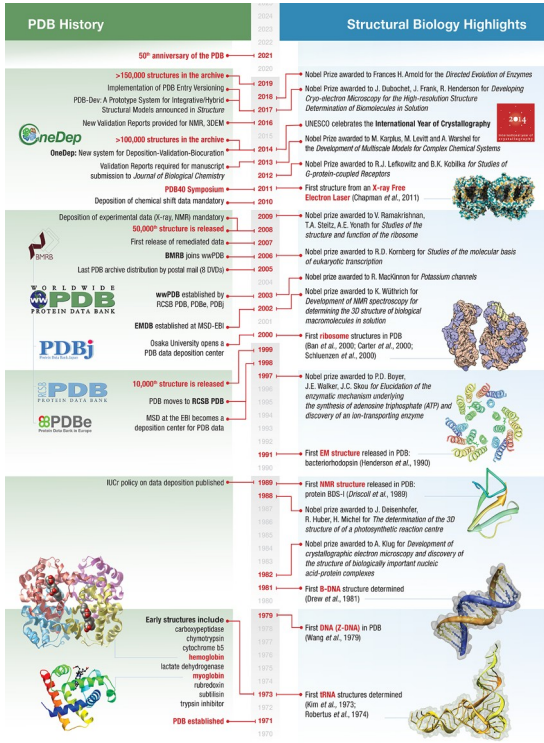


- **1966:** Cyrus Levinthal, et. al. publish paper on the use of molecular graphics and computer simulation (Levinthal, C.; Scientific American, 1966, 214: 42).
- **1969:** Levitt and Lifson report the use of force fields to refine protein conformations derived from experimental data (Michael Levitt and Shneior Lifson; J. Mol. Biol., 1969, 46, 269-279).

  Ken Thompson, Dennis Ritchie and Joseph Ossanna develop a new operating system, UNIX, for the DEC PDP-7.

  Dennis Ritchie and Brian Kernighan create “C” at Bell Labs.
- **1971:** Rahman and Stillinger–MD simulation of water.
- **1973:** The Brookhaven Protein Data Bank is announced (Acta. Cryst. B, 1973, 29: 1746);

  N.L. Allinger describes the modeling of hydrocarbons with a new force field, MM1 (Allinger, N.L.; Sprague, J.T.; J. Amer. Chem. Soc., 1973, 95: 3893).



History and evolution of Computational Chemistry

- **1975:** Microsoft is founded by Bill Gates and Paul Allen.
- **1977:** Martin Karplus, et. al. publish the first molecular dynamics study of a protein (McCammon, J.A.; Gelin, B.R.; Karplus, M.; Nature, 1977 267: 585-590).
- **1980:** The first issue of the Journal of Computational Chemistry is published.
- **1981:** AMBER Force-field for proteins/DNA calculations; IBM introduces its Personal Computer to the market.
- **1982:** An algorithm for docking small molecules to receptors (later to become the DOCK program) is published by Irwin Kuntz and colleagues (Kuntz, I.D.; Blaney, J.M.; Oatley, S.J.; Langridge, R.; Ferrin, T.E.; J. Mol. Biol., 1982, 161, 269).

Dynamics of folded proteins

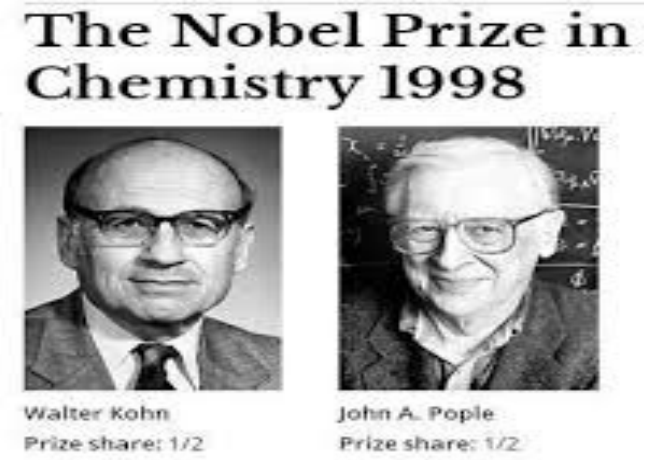
J. Andrew McCammon, Bruce R. Gelin & Martin Karplus

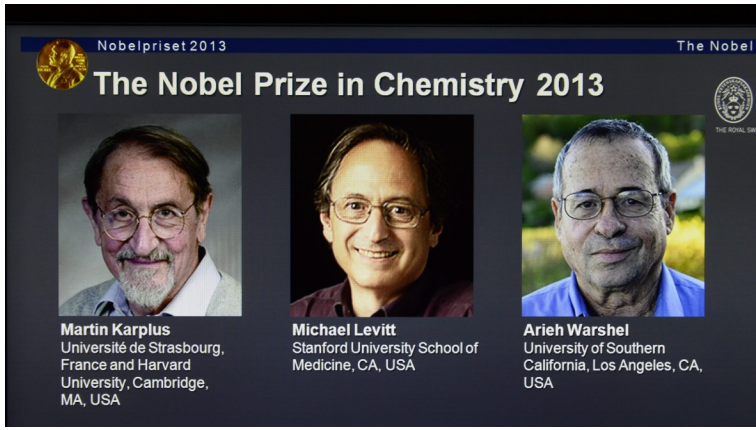
Department of Chemistry, Harvard University, Cambridge, Massachusetts 02138

The dynamics of a folded globular protein (bovine pancreatic trypsin inhibitor) have been studied by solving the equations of motion for the atoms with an empirical potential energy function. The results provide the magnitude, correlations and decay of fluctuations about the average structure. These suggest that the protein interior is fluid-like in that the local atom motions have a diffusional character.

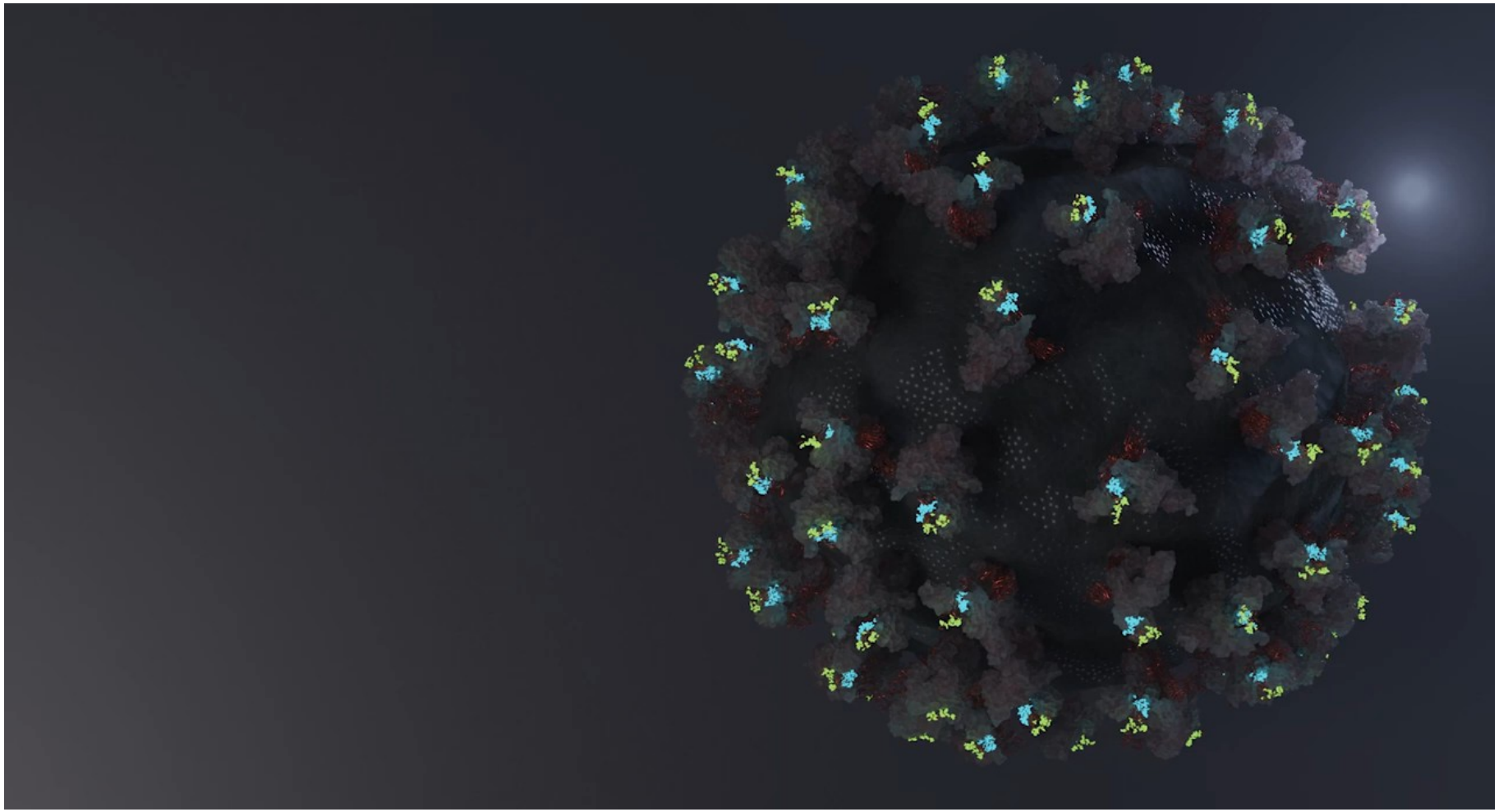
History and evolution of Computational Chemistry

- **1983:** Martin Karplus, et. al. publish description of the **CHARMM** program (Brooks, B.R.; Bruccoleri, R.E.; Olafson, B.D.; States, D.J., Swaminathan, S. and Karplus, M.; J. Comp. Chem., 1983, 4:187-217); W. F. van Gunsteren, et. al. publish description of molecular dynamics of proteins using **GROMOS** program (van Gunsteren, W. F.; Berendsen, H. J. C.; Hermans, J.; Hol, W. G. J.; Postma, J. P. M.; Proc. Natl. Acad. Sci., 1983, 80: 4315); The **Compact Disk (CD)** is launched.
- **1983 – Till now:** A lot of developments!! (Not mentioned here.)
- **NOBLE PRIZES:**
- **1998: Walter Kohn**, "for his development of the density-functional theory", and **John Pople**, "for his development of computational methods in quantum chemistry", received the 1998 Nobel Prize in Chemistry.
- **2013: Martin Karplus, Michael Levitt and Arieh Warshel** received the 2013 Nobel Prize in Chemistry for "the development of multiscale models for complex chemical systems".





Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2)



Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2)

nature chemistry

ARTICLES

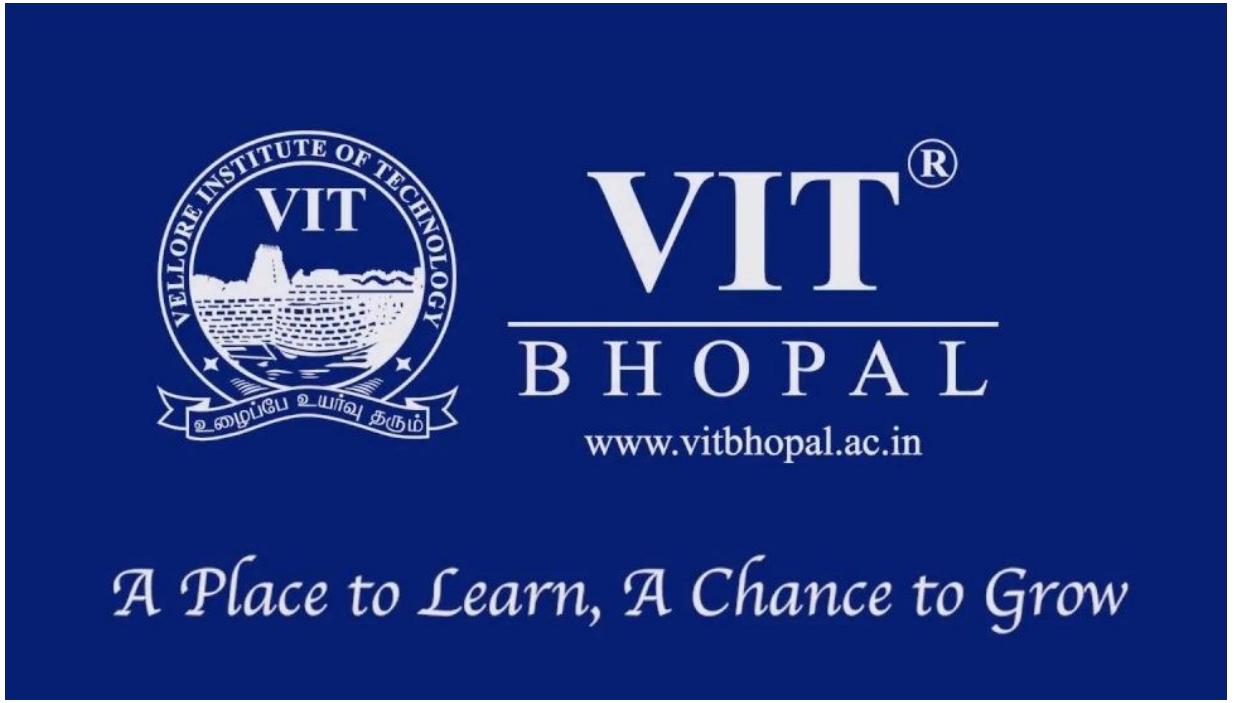
https://doi.org/10.1038/s41557-021-00707-0

Check for updates

SARS-CoV-2 simulations go exascale to predict dramatic spike opening and cryptic pockets across the proteome

Maxwell I. Zimmerman[1,2], Justin R. Porter[1,2], Michael D. Ward[1,2], Sukrit Singh[1,2], Neha Vithani[1,2], Artur Meller[1,2], Upasana L. Mallimadugula[1,2], Catherine E. Kuhn[1,2], Jonathan H. Borowsky[1,2], Rafal P. Wiewiora[3,4], Matthew F. D. Hurley[5], Aoife M. Harbison[6], Carl A. Fogarty[6], Joseph E. Coffland[7], Elisa Fadda[6], Vincent A. Voelz[5], John D. Chodera[4] and Gregory R. Bowman[1,2]

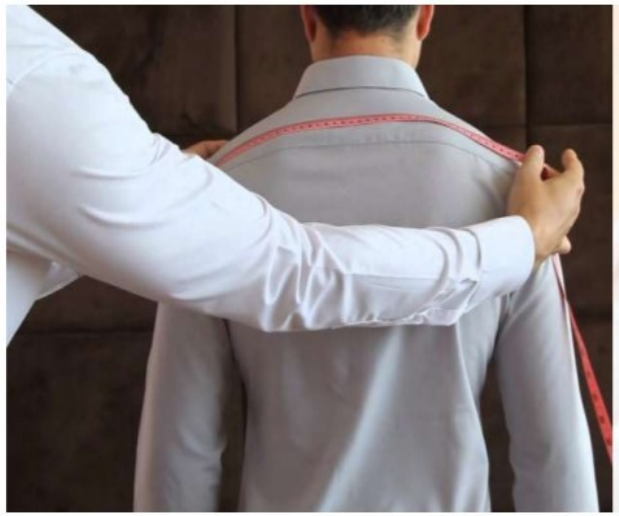
SARS-CoV-2 has intricate mechanisms for initiating infection, immune evasion/suppression and replication that depend on the structure and dynamics of its constituent proteins. Many protein structures have been solved, but far less is known about their relevant conformational changes. To address this challenge, over a million citizen scientists banded together through the Folding@home distributed computing project to create the first exascale computer and simulate 0.1 seconds of the viral proteome. Our adaptive sampling simulations predict dramatic opening of the apo spike complex, far beyond that seen experimentally, explaining and predicting the existence of 'cryptic' epitopes. Different spike variants modulate the probabilities of open versus closed structures, balancing receptor binding and immune evasion. We also discover dramatic conformational changes across the proteome, which reveal over 50 'cryptic' pockets that expand targeting options for the design of antivirals. All data and models are freely available online, providing a quantitative structural atlas.



Introduction: Measurements

Dr. Saurav Prasad

Measurements in everyday life



Measurement of length

Measurement of temperature

Measurement of mass

Measurement of volume

Physical quantity

A physical property that can be measured and described by a number is called physical quantity.

Examples:

- Mass of a person is 65 kg.
- Length of a table is 3 m.
- Area of a hall is 100 m^2.
- Temperature of a room is 300 K

Types of physical quantities

1. Fundamental quantities:

The physical quantities which do not depend on any other physical quantities for their measurements are known as fundamental quantities.

Examples:

- Mass
- Length
- Time
- Temperature

2. Derived quantities:

The physical quantities which depend on one or more fundamental quantities for their measurements are known as derived quantities.

Examples:

- Area
- Volume
- Speed
- Force

Units of measurement

The standard used for the measurement of a physical quantity is called a unit.

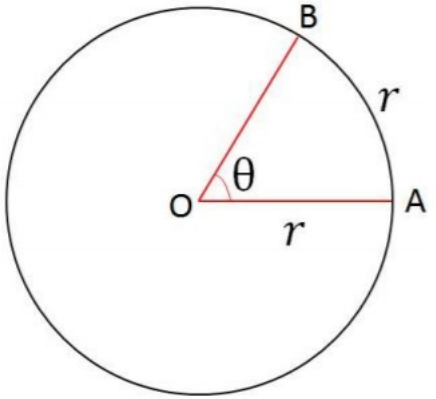
Examples:

- metre, foot, inch for length
- kilogram, pound for mass
- second, minute, hour for time
- fahrenheit, kelvin for temperature

Seven fundamental & two supplementary units

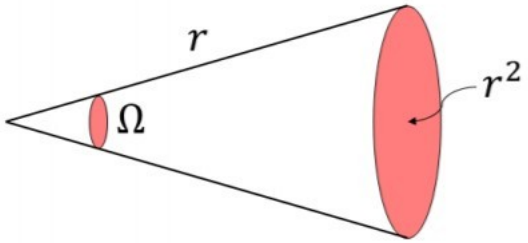
FUNDAMENTAL QUANTITY	SI UNIT	SYMBOL
Length	metre	m
Mass	kilogram	kg
Time	second	s
Temperature	kelvin	K
Electric current	ampere	A
Luminous intensity	candela	cd
Amount of substance	mole	mol

1. Radian: It is used to measure plane angle



θ = 1 radian

2. Steradian: It is used to measure solid angle



Ω = 1 steradian

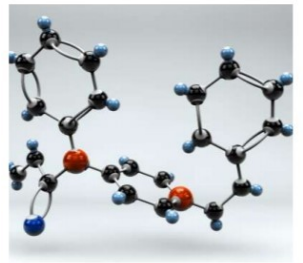
SI prefixes and some practical units for measuring length

Factor	Name	Symbol	Factor	Name	Symbol
10^{24}	yotta	Y	10^{-1}	deci	d
10^{21}	zetta	Z	10^{-2}	centi	c
10^{18}	exa	E	10^{-3}	milli	m
10^{15}	peta	P	10^{-6}	micro	μ
10^{12}	tera	T	10^{-9}	nano	n
10^{9}	giga	G	10^{-12}	pico	p
10^{6}	mega	M	10^{-15}	femto	f
10^{3}	kilo	k	10^{-18}	atto	a
10^{2}	hecto	h	10^{-21}	zepto	z
10^{1}	deka	da	10^{-24}	yocto	y

1 micron = 10^{-6} m

Bacterias

1 nanometer = 10^{-9} m



Molecules

- Astronomical unit = It is defined as the mean distance of the earth from the sun.
- 1 astronomical unit = 1.5×10^{11} m

Distance of planets

Dimensions of a physical quantity

The powers of fundamental quantities in a derived quantity are called dimensions of that quantity.

Fundamental quantities	Dimensions
Length	[L]
Mass	[M]
Time	[T]
Temperature	[K]
Current	[A]
Amount of substance	[N]
Luminous intensity	[J]

Example:

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

$$= \frac{\text{Mass}}{\text{length} \times \text{breath} \times \text{height}}$$

$$[\text{Density}] = \frac{[\text{M}]}{[\text{L}] \times [\text{L}] \times [\text{L}]} = \frac{[\text{M}]}{[\text{L}^3]} = [\text{ML}^{-3}]$$

Hence the dimensions of density are 1 in mass and -3 in length.

To check the correctness of equation

To convert units

To derive a formula

Question

1. The velocity 'v' of a particle depends upon the time 't' according to the equation

$$v = \sqrt{ab} + bt^3 + \frac{c}{d + t^2}$$

Determine the units of a, b, c and d. All have SI units

Question

2. The period *'T'* of a simple pendulum is measured in time units and is described by

$$T = 2\pi\sqrt{\frac{l}{g}}$$

Where *l* is the length of the pendulum and 'g' is the free fall acceleration in units of length divided by square of time. Show that this equation is dimensionally correct.

Question

3. The force F acting on a body moving in a circular path depends on mass of the body (m), velocity (v) and radius (r) of the circular path. Obtain the expression for the force by dimensional analysis method. (Take the value of k=1)

Accuracy and precision of measurement

It refers to the closeness of a measurement to the true value of the physical quantity.

Example:

- True value of mass = 25.67 kg
- Mass measured by student A = 25.61 kg
- Mass measured by student B = 25.65 kg
- The measurement made by student B is more accurate.

It refers to the limit to which a physical quantity is measured.

Example:

- Time measured by student A = 3.6 s
- Time measured by student B = 3.69 s
- Time measured by student C = 3.695 s
- The measurement made by student C is most precise.

Significant figures

The total number of digits (reliable digits + last uncertain digit) which are directly obtained from a particular measurement are called significant figures.

Mass = 6.11 g

3 significant figures

Speed = 67 km/h

2 significant figures

Rules for counting significant figures

1

All non-zero digits are significant.

Number	Significant figures
16	2
35.6	3
6438	4

2

Zeros between non-zero digits are significant.

Number	Significant figures
2<u>0</u>5	3
3<u>00</u>8	4
6<u>0</u>.005	5

3

Terminal zeros in a number without decimal are not significant unless specified by a least count.

Number	Significant figures
4<u>00</u>	1
305<u>0</u>	3
(20 ± 1) s	2

4

Terminal zeros that are also to the right of a decimal point in a number are significant.

Number	Significant figures
64.<u>00</u>	4
3.6<u>0</u>	3
25.06<u>0</u>	5

5

If the number is less than 1, all zeroes before the first non-zero digit are not significant.

Number	Significant figures
<u>0.00</u>64	2
<u>0.0</u>850	3
<u>0.000</u>2050	4

6

During conversion of units use powers of 10 to avoid confusion.

Number	Significant figures
2.700 m	4
2.700×10^{2} cm	4
2.700×10^{-3} km	4

Exact numbers

- Exact numbers are either defined numbers or the result of a count.
- They have infinite number of significant figures because they are reliable.

By definition	By counting
1 dozen = 12 objects	45 students
1 hour = 60 minute	5 apples
1 inch = 2.54 cm	6 faces of cube

Rules for rounding off a measurement

1

If the digit to be dropped is less than 5, then the preceding digit is left unchanged.

Number	Round off up to 3 digits
64.62	64.6
3.651	3.65
546.3	546

2

If the digit to be dropped is more than 5, then the preceding digit is raised by one.

Number	Round off up to 3 digits
3.479	3.48
93.46	93.5
683.7	684

3

If the digit to be dropped is 5 followed by digits other than zero, then the preceding digit is raised by one.

Number	Round off up to 3 digits
62.354	62.4
9.6552	9.66
589.51	590

4

If the digit to be dropped is 5 followed by zero or nothing, the last remaining digit is increased by 1 if it is odd, but left as it is if even.

Number	Round off up to 3 digits
53.350	53.4
9.455	9.46
782.5	782

Significant figures in calculations

Addition & subtraction

The final result would round to the same decimal place as the least precise number.

Example:

- 13.2 + 34.654 + 59.53 = 107.384 = 107.4
- 19 – 1.567 - 14.6 = 2.833 = 3

Multiplication & division

The final result would round to the same number of significant digits as the least accurate number.

Example:

- 1.5 x 3.67 x 2.986 = 16.4379 = 16
- 6.579/4.56 = 1.508 = 1.51

Questions

1. How many Significant figures in each term?
   a. 34.6209 b. 0.003048 c. 5010.0 d. 4032.090
2. Solve the following equations using the correct number of significant figures.

   a. 0.003 + 3.5198 + 0.0118 b. 6.90 / 2.8952

   c. 98.1 x 0.03

3. A meter has 1000 millimeters in it. How many significant figures does this number have?

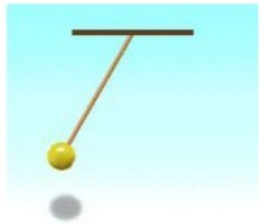
Errors in measurement

Difference between the actual value of a quantity and the value obtained by a measurement is called an error.

Error = |actual value – measured value|

Error analysis

For example, suppose you measure the oscillation period of a pendulum with a stopwatch five times.



Trial no (i)	1	2	3	4	5
Measured value (X_i)	3.9	3.5	3.6	3.7	3.5

The magnitude of the difference between mean value and each individual value is called absolute error.

$$\Delta X_i = |\bar{X} - X_i|$$

The absolute error in each individual reading:

X_i	3.9	3.5	3.6	3.7	3.5
ΔX_i	0.3	0.1	0	0.1	0.1

The average of all the five readings gives the most probable value for time period.

$$\bar{X} = \frac{1}{n} \sum X_i$$

$$\bar{X} = \frac{3.9 + 3.5 + 3.6 + 3.7 + 3.5}{5} = \frac{18.2}{5}$$

$$\bar{X} = 3.64 = 3.6$$

The arithmetic mean of all the absolute errors is called mean absolute error.

$$\Delta\bar{X} = \frac{1}{n} \sum \Delta X_i$$

$$\Delta\bar{X} = \frac{0.3 + 0.1 + 0 + 0.1 + 0.1}{5} = \frac{0.6}{5}$$

$$\Delta\bar{X} = 0.12 = 0.1$$

Error analysis

- The most common way adopted by scientist and engineers to report a result is:

$$\text{Result} = \text{best estimate} \pm \text{error}$$

- It represent a range of values and from that we expect a true value fall within.
- Thus, the period of oscillation is likely to be within (3.6 ± 0.1) s.

The relative error multiplied by 100 is called as percentage error.

$$\text{percentage error} = \text{relative error} \times 100$$

$$\text{percentage error} = 0.028 \times 100$$

$$\text{percentage error} = 2.8\ \%$$

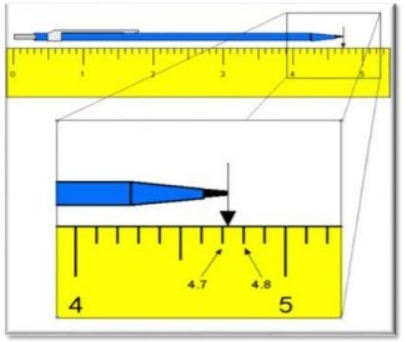
The relative error is defined as the ratio of the mean absolute error to the mean value.

$$\text{relative error} = \Delta\bar{X} / \bar{X}$$

$$\Delta\bar{X} / \bar{X} = \frac{0.1}{3.6} = 0.0277$$

$$\Delta\bar{X} / \bar{X} = 0.028$$

Least count error is the error associated with the resolution of the instrument.



- The least count error of any instrument is equal to its resolution.
- Thus, the length of pen is likely to be within (4.7 ± 0.1) cm.

Combination of errors

In different mathematical operations like addition, subtraction, multiplication and division the errors are combined according to some rules.

- Let ΔA be absolute error in measurement of A
- Let ΔB be absolute error in measurement of B
- Let ΔX be absolute error in measurement of X

When $X = A \pm B$

$$\frac{\Delta X}{X} = \frac{\Delta A + \Delta B}{A \pm B}$$

$$\Delta X = \Delta A + \Delta B$$

When $X = A \times B$ or A / B

$$\frac{\Delta X}{X} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

$$\Delta X = \left(\frac{\Delta A}{A} + \frac{\Delta B}{B}\right) X$$

When $X = A^n$

$$\frac{\Delta X}{X} = n\frac{\Delta A}{A}$$

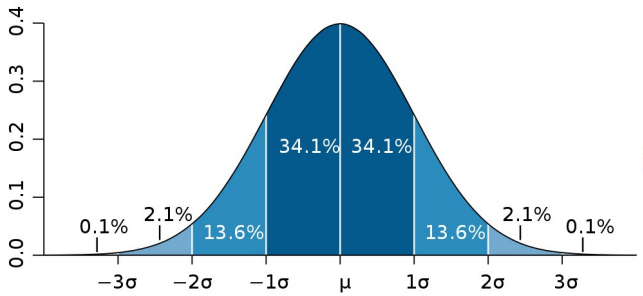
$$\Delta X = n\frac{\Delta A}{A} X$$

Standard deviation and standard error of the mean

- The standard deviation is a statistic that measures the dispersion of a dataset relative to its mean and is calculated as the square root of the variance.

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

σ = population standard deviation
N = the size of the population
x_i = each value from the population
μ = the population mean



The standard error of the mean (SEM) can be expressed as:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

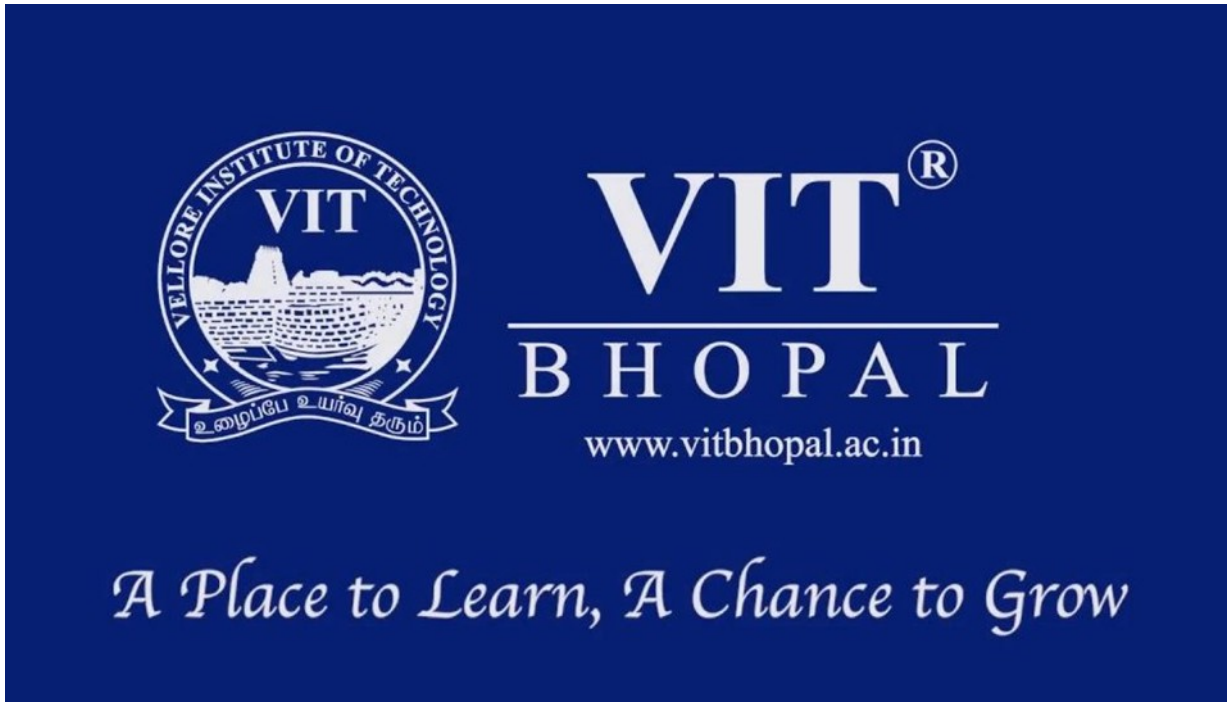
where

σ is the standard deviation of the population.

n is the size (number of observations) of the sample.

Question

1) The marks of a class of eight students (that is, a statistical population) are the following eight values: 2, 4, 4, 4, 5, 5, 7, 9. Calculate mean, absolute error, mean absolute error, relative error, percentage error, variance, standard deviation, and standard error.
2) 5.74 g of a substance occupies 1.2 cm^3. Express its density by keeping the significant figures in view.
3) The radius of a circle is measured to be 10.5 ± 0.2 m. Calculate the (a) area and (b) circumference of the circle and give the uncertainty in each value.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Thank you